

LiDAR Inertial Mapping of Non-Manhattan Topologies for UAV-Based Aircraft Inspection in GNSS-Denied Environments

R. Pugliese * D. Abel *

** Institute of Automatic Control, RWTH Aachen University
Campus-Boulevard 30, 52074 Aachen
e-mail: r.pugliese@irt.rwth-aachen.de*

Abstract: In this paper we propose and evaluate a high resolution intensity yielding adaption of a 3D mapping approach for airplane inspection by autonomous unmanned aerial vehicles. We solely use a 16 column laserscanner and an industrial-grade inertial measurement unit, without a priori knowledge of the inspection object dimensions or its surroundings. Our approach includes the introduction of a novel cylinder feature detection and an according modification of the optimization-based state estimation. Robustness of the point cloud registration is proven based on measurements in a GNSS-denied hangar environment with a Boeing 737-500, comprising only sparse plane and edge features and for highly dynamic movement involving rapid roll or pitch rotation. The approach is suitable for the given application to datasets with strong non-Manhattan characteristics, as opposed to several state of the art algorithms. By means of including the intensity information, 2D projections of the 3D registration result depict a remarkable level of detail and sharpness, resembling that of camera images. Thus allowing for complete coverage of the environment with high spatial resolution using the discrete points and voxel grids with small leaf sizes, the obtained results perspectively can be used for visually matching multi-channel camera images. By fusion of such visual information with the LiDAR generated projections and using the LiDAR inertial odometry, 3D defect maps of the aircraft surface can be constructed, enabling the automated localization of structural damages that are detectable using convolutional neural networks, as shown in previous publications.

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1. INTRODUCTION

Operation of civil aircraft requires mandatory regular inspection in order to detect damages to the surface structures that could lead to safety risks in the long term if not remedied. Such defects occur predominantly as a result of lightning strikes that are conducted along the surface, leaving material damage, sometimes the size of a pinhead, at the points of impact and discharge. Manual execution of the inspection task by specifically trained staff requires the usage of bosun's chairs suspended from ropes and consequently is a time consuming process, causing long periods of unavailability of the aircraft and thus high costs for the airlines. Autonomous UAV offer great potential for automating the task and significantly reducing the downtimes. Perspectively, the inspection task could be shifted to the runway and performed directly after landing, eliminating the need for hangar visits between flights in case no maintenance is required.

Irregularities, i.e. potential defects of the aircraft surface, can be detected automatically using CNN feature detection on camera images, see Konrad et al. (2019). However, due to the large dimensions of the aircraft surface, global

localization of these defect candidates is necessary for easy spotting and fast manual follow-up examination of the areas depicted in images that were classified as defect positive. Ideally, the creation of a 3D digital twin of the aircraft is desired for this task. Considering the great variety of airplane designs, the data-based approach of incorporating a priori knowledge of the object geometry should be avoided in the process. Besides, the chosen approach should preferably work independently of external hardware installations, which would prove difficult in crowded hangar environments or even more so, on the runway.

For 3D environmental perception and remote sensing, there are multiple sensor technology options, such as stereo cameras (Engel et al. (2015)) or radar (Marck et al. (2013)). However, stereo cameras are passive sensors, thus being dependent on the illumination conditions and therefore rely on external light sources. Additionally, the camera mounted on the given UAV can take high resolution images but only at specified triggering times and offers no continuous data stream. Radar on the other hand commonly suffers from low spatial resolution and a limited field of view. Our approach uses a Velodyne VLP 16 Puck

laserscanner that was already used for object detection, collision avoidance (Pugliese et al. (2019)) and distance measurement (Pugliese et al. (2021)) for the given application in previous publications. Due to the large scale of the aircraft and the typically low working distances during the inspection, individual LiDAR sweeps may only depict small sections of the overall surface, therefore requiring a mapping approach to register the frames, i.e. estimate the rigid transformation $T_a^b \in SE(3)$ from LiDAR frame L_a to frame L_b , consisting of a rotation $R_a^b \in SO(3)$ and a translation $p_a^b \in \mathbb{R}^3$ given by the sensor movement, in order to generate a coherent and complete 3D model.

In recent years, significant advancements have been made on the field of LiDAR-based simultaneous localization and mapping (SLAM). Particle filter based approaches (Nie et al. (2020)) may reach high accuracy but their performance heavily depends on the number of particles. In spite of their inherent suitability for parallelization, for most 6-DoF problems the computational costs exceed the available onboard hardware capabilities of mobile robots by far, restricting the applicability of Monte-Carlo algorithms primarily to cases in which 2D mapping suffices. Zhang and Singh (2014) proposed LOAM, a LiDAR-only odometry and mapping approach to which IMU data can be added optionally, but solely for scan deskewing and not for probabilistic sensor fusion. The algorithm is based on plane and edge feature detection, yielding excellent results in Manhattan datasets, i.e. architectures with perpendicular a ground plans but lacks robustness in environments comprising large homogeneous curved surfaces such as aircraft fuselages and for dynamic roll or pitch movement. LIOM (Ye et al. (2019)) extends the approach to a tightly coupled LiDAR inertial sensor fusion and is used as a basis for this work but it is based on the same feature detection notion as LOAM, also exhibiting the same weakness. In fact, the authors reject edge features, stating they cannot improve the odometry results, which is arguably not universally applicable to all datasets. LIO-SAM (Shan et al. (2020)) uses a factor graph representation, incorporating GNSS measurements and loop-closing, while the LiDAR feature detection remains unchanged. Kumar et al. (2017) propose a LiDAR inertial mapping and localization approach for UAV-based pipeline inspection, which requires the perpendicular attachment of two LiDAR sensors that are fused using Kalman filtering. However, for aircraft inspection this approach is not applicable as the environment comprises only few dominant cylinders that are parallel and lack angles, namely the fuselage and the engines, thus undermining global complete observability of the UAV pose without the usage of additional features.

In this publication we propose a novel modification and extension of LIOM to make it suitable for the application to datasets with strong non-Manhattan characteristics and for the given use case in particular. More precisely, we introduce cylinder feature detection in order to make the conventional plane and edge features robust to homogeneous curved surfaces such as aircraft fuselages. This involves the computer vision part as well as an adaption of the cost function of the optimization problem that is employed for pose and state estimation. In section 2, the methodology and the adaptations of individual modules of the framework are presented. The mapping results are then

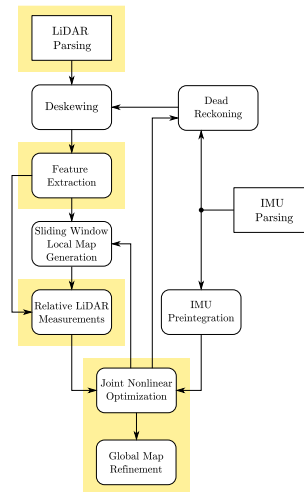


Fig. 1. Schematic overview of the mapping framework based on LIOM (Ye et al. (2019)). Modules with proposed modifications are highlighted.

given and evaluated based on measurements in a hangar environment using an external indoor positioning system (iGPS) as reference in section 3.

2. METHODOLOGY

Fig. 1 shows a flow chart of the chosen mapping approach according to LIOM. Yellow highlighting of individual modules indicates proposed modifications or extensions. In the following, these adaptations are discussed in the corresponding subsections. For a detailed explanation of the overall architecture we refer the reader to Ye et al. (2019) as a thorough description of the framework must be omitted due to space limitations.

2.1 LiDAR parsing

While not part of the actual SLAM framework, the source built ROS (Stanford Artificial Intelligence Laboratory et al. (2018)) Velodyne driver is modified to allow parsing of dual returns. Due to beam divergence, the light emitted from one of the 16 semiconductor laser diodes may hit more than one object in a specific firing interval. Dual return mode not only registers the strongest return, i.e. the one with the highest intensity at the IR detector, but also the latest return in case it differs from the strongest one. This implicates that the UDP packets arriving from the sensor have to be parsed at a doubled packet rate of 1508 Hz instead of the common 754 Hz. This allows obtaining higher resolution point clouds without increasing measurement duration at the cost of slower processing. Note, that rotation frequency in contrast does not affect data rate. This parameter is usually set to 600 rpm as it allows for higher azimuthal resolution of individual scans compared to the maximum value of 1200 rpm. For capturing the high movement dynamics in our practical evaluation (see section 3), rotation speed was set to 1200 rpm. Nevertheless, we obtain high resolution mapping results.

2.2 Feature Extraction

As a basis, the feature detection from LOAM is used. For each point x_n in each of the 16 Scans $\mathcal{S}_i, i \in \{1, \dots, 16\}$

contained in a sweep $\mathcal{P}$, a heuristic expression for the local azimuthal curvature $c_{i,n}$ is calculated according to

$$c_{i,n} = \frac{1}{|\mathcal{S}_i| \cdot \|\mathbf{x}_n\|} \left\| \sum_{m \in \mathcal{S}_i, m \neq n} (\mathbf{x}_n - \mathbf{x}_m) \right\|. \quad (1)$$

The points with minimum curvature values below a pre-defined threshold are selected as planar points, while the points with maximum values above a fixed threshold are considered to be edge points. As long as some additional well-posed boundary conditions limiting for example the maximum number of extracted points per quadrant are fulfilled, these planar and edge points are extracted as features. Evidently, horizontal edges cannot be detected using this approach and homogeneous curved surfaces such as aircraft fuselages also prove to be problematic. Hence, the addition of global information about curved surface geometry to the approach of using individual feature points is a reasonable idea to overcome these deficits. As shown previously, the combination of cylinder RANSAC and intensity-based prefiltering can be used for aircraft fuselage segmentation (Pugliese et al. (2021)). Thus, we propose the calculation of cylinder features according to algorithm 1, which is implemented using PCL (Rusu and Cousins (2011)).

Algorithm 1 Cylinder feature extraction

Input: point cloud of current sweep $\mathcal{P}_k$
Output: cylinder cap centers $\mathbf{c}_{1,i}$, $\mathbf{c}_{2,i}$ and radii r_i , $i \in 1, \dots, \text{maxcylindercount}$

```

1: begin
2:   initialize empty passthrough point cloud  $\mathcal{P}_{k,\text{filtered}}$ ;
3:   for each point  $\mathbf{x}_n \in \mathcal{P}_k$  do
4:     if  $\text{intensity}(\mathbf{x}_n) < \text{int}_{\max} \wedge \text{intensity}(\mathbf{x}_n) >$   

 $\text{int}_{\min}$  then
5:        $\mathcal{P}_{k,\text{filtered}} \leftarrow \mathcal{P}_{k,\text{filtered}} + \mathbf{x}_n$ ;
6:     end if
7:   end for
8:   build kd-tree on  $\mathcal{P}_{k,\text{filtered}}$ ;
9:   for each point  $\mathbf{x}_j \in \mathcal{P}_{k,\text{filtered}}$  do
10:    perform KNN search around  $\mathbf{x}_j$ ;
11:    perform least squares surface normal estimation  

    at query point  $\mathbf{x}_j$  using Principal Component Analysis;
12:  end for
13:   $i = 1$ ;
14:  while  $i \leq \text{maxcylindercount}$  do
15:    perform cylinder RANSAC on  $\mathcal{P}_{k,\text{filtered}}$ ;
16:    if  $|\mathcal{P}_{k,i,\text{inliers}}| / |\mathcal{P}_k| > \text{threshold}$  then
17:       $\mathbf{c}_{1,i}, \mathbf{c}_{2,i}, r_i \leftarrow \text{estimatedmodelcoefficients}$ ;
18:       $\mathcal{P}_{k,\text{filtered}} \leftarrow \mathcal{P}_{k,\text{filtered}} - \mathcal{P}_{k,i,\text{inliers}}$ ;
19:       $i \leftarrow i + 1$ ;
20:    else
21:      break;
22:    end if
23:  end while
24: end
```

2.3 Relative LiDAR measurements

In order to increase the feature density and to facilitate finding correspondences, the features within a fixed size sliding window from the last timestamps $\{o, \dots, p, \dots, i\}$ are

agglomerated in the pivot frame of sweep $\mathcal{P}_p$ using the transformations previously estimated by the optimization. The relative LiDAR measurements define the relation between the features from all frames subsequent to the pivot frame at timestamps $\alpha \in \{p+1, \dots, i+1\}$, including the newly arriving current LiDAR sweep $\mathcal{P}_{i+1}$, to the features in the local map. For the plane features the relative LiDAR measurements $\mathbf{m}_{\text{plane},\alpha} = [\mathbf{x}_{\text{plane},L_\alpha}^{L_\alpha}, \mathbf{n}^{L_p}, d]$ are

$$\mathbf{m}_{\text{edge},\alpha} = [\mathbf{x}_{\text{edge},L_\alpha}^{L_\alpha}, \mathbf{l}^{L_p}, \mathbf{x}_0^{L_p}] \quad (2)$$

with the unit direction vector of the line segment $\mathbf{l}^{L_p}$ in the local map frame and a position vector of a point on that line segment $\mathbf{x}_0^{L_p}$.

For the introduced cylinder features, the model coefficients are yielded directly by algorithm 1, rendering parameter estimation by further KNN-search futile. As opposed to the relative plane and edge measurements, cylinder features do not contain measured feature points from the original point cloud, but the cylinder cap centers, which can be considered virtual measurements. Due to the unknown dilatation of the cylinders, correspondences between a cylinder with radius r_a from a frame L_α and the cylinders in the local map with radii $r_{p,i}$, $i \in \{1, \dots, \text{maxcylindercount}\}$ are not established based on spatial proximity of the cap centers but via $\arg \min_i |r_{p,i} - r_a|$ with the constraint $|r_{p,i} - r_a| < r_{\text{threshold}}$. We define the relative LiDAR measurement as

$$\mathbf{m}_{\text{cyl},\alpha} = [\mathbf{c}_{1,L_\alpha}^{L_\alpha}, \mathbf{c}_{2,L_\alpha}^{L_\alpha}, \mathbf{c}_{1,i}^{L_p}, \mathbf{c}_{2,i}^{L_p}]. \quad (3)$$

2.4 Joint Nonlinear Optimization

The aim of the joint nonlinear optimization is to find a consensus between the LiDAR and inertial predicted transformations and to estimate the 16-dimensional state vector

$$\mathbf{X}_{B_k}^W = [\mathbf{p}_{B_k}^W, \mathbf{v}_{B_k}^W, \mathbf{q}_{B_k}^W, \mathbf{b}_{a_k}^T, \mathbf{b}_{\omega_k}^T]^T,$$

containing the position $\mathbf{p}_{B_k}^W$, velocity $\mathbf{v}_{B_k}^W$ and orientation described as unit quaternion $\mathbf{q}_{B_k}^W$ of the IMU body frame B_k at timestep k with respect to the world frame W that is used for constructing the global map. The IMU biases of accelerometer $\mathbf{b}_{a_k}$ and gyroscope $\mathbf{b}_{\omega_k}$ are estimated as well. The pose of the body frame w.r.t. the world frame $\mathbf{T}_{B_k}^W$ also yields the pose of the LiDAR frame $\mathbf{T}_{L_k}^W$ by using the extrinsic parameter $\mathbf{T}_B^L$ that can be specified as a fixed value in case the parameter is known or estimated as well. Also, not only the current state is estimated but all states within the optimization window, i.e. all last states $\mathbf{X}_{B_k}^W$, $k \in \{p, \dots, i+1\}$ since the timestamp of the pivot frame, which is known as fixed-lag smoothing.

As the features were modified previously, the residuals in the cost function that is minimized for finding the

maximum a posteriori probability state estimation have to be adjusted as well:

$$\arg \min_{\mathbf{X}} \frac{1}{2} \left\{ \|\mathbf{r}_{\mathcal{P}}(\mathbf{X})\|^2 + \sum_{\alpha=p+1}^{i+1} \left(\sum_{\mathbf{m}_{\text{plane},\alpha}} \|\mathbf{r}_{\text{plane}}(\mathbf{m}_{\text{plane},\alpha}, \mathbf{X})\|_{C_{L_\alpha}^m}^2 + \sum_{\mathbf{m}_{\text{edge},\alpha}} \|\mathbf{r}_{\text{edge}}(\mathbf{m}_{\text{edge},\alpha}, \mathbf{X})\|_{C_{L_\alpha}^m}^2 + \sum_{\mathbf{m}_{\text{cyl},\alpha}} \|\mathbf{r}_{\text{cyl}}(\mathbf{m}_{\text{cyl},\alpha}, \mathbf{X})\|_{C_{L_\alpha}^m}^2 \right) + \sum_{\beta=p}^i \|\mathbf{r}_{\mathcal{B}}(z_{\beta+1}^\beta, \mathbf{X})\|_{C_{B_{\beta+1}}^{B_\beta}}^2 \right\} \quad (4)$$

with the unmodified priors of the marginalization $\mathbf{r}_{\mathcal{P}}$, residuals of the IMU preintegration $\mathbf{r}_{\mathcal{B}}$ and residuals of the plane feature correspondences $\mathbf{r}_{\text{plane}}$ (point-to-plane distances). The residuals of the edge feature correspondences can be derived accordingly as point-to-line distances:

$$\mathbf{r}_{\text{edge}}(\mathbf{m}_{\text{edge},\alpha}, \mathbf{X}) = \|\mathbf{l}^{L_p} \times (\mathbf{x}_0^{L_p} - \mathbf{R}_{L_\alpha}^{L_p} \mathbf{x}_{\text{edge},L_\alpha}^{L_\alpha} + \mathbf{p}_{L_\alpha}^{L_p})\| \quad (5)$$

For cylinder features we choose a similar approach of using the sum of the cap center distances to the cylinder axis in the pivot frame as residuals since the radii were already taken into account when establishing the correspondences:

$$\mathbf{r}_{\text{cyl}}(\mathbf{m}_{\text{cyl},\alpha}, \mathbf{X}) = \frac{1}{\|\mathbf{c}_{2,i}^{L_p} - \mathbf{c}_{1,i}^{L_p}\|} \cdot \left(\left\| (\mathbf{c}_{2,i}^{L_p} - \mathbf{c}_{1,i}^{L_p}) \times (\mathbf{c}_{1,i}^{L_p} - \mathbf{R}_{L_\alpha}^{L_p} \mathbf{c}_{1,L_\alpha}^{L_\alpha} + \mathbf{p}_{L_\alpha}^{L_p}) \right\| + \left\| (\mathbf{c}_{2,i}^{L_p} - \mathbf{c}_{1,i}^{L_p}) \times (\mathbf{c}_{1,i}^{L_p} - \mathbf{R}_{L_\alpha}^{L_p} \mathbf{c}_{2,L_\alpha}^{L_\alpha} + \mathbf{p}_{L_\alpha}^{L_p}) \right\| \right). \quad (6)$$

The optimization problem is solved using Gauss-Newton algorithm as implemented in Ceres solver (Agarwal et al. (2022)).

2.5 Global Map Refinement

While in the previous section the states were optimized in a sliding window, global map refinement properly aligns the current features with the world map using a second optimization that is running asynchronously in the back-end. It is formulated as a second optimization problem, for which the odometry serves first as a prior and its rotational component is then used as a rotational constraint. This optimization problem accordingly takes the form

$$\arg \min_{\mathbf{T}_{L_k}^W} \sum_{\mathbf{m}_{\text{plane},k}} \left\| \mathbf{r}_{\text{plane}}(\mathbf{m}_{\text{plane},k}, \mathbf{T}_{L_k}^W) \right\|^2 + \sum_{\mathbf{m}_{\text{edge},k}} \left\| \mathbf{r}_{\text{edge}}(\mathbf{m}_{\text{edge},k}, \mathbf{T}_{L_k}^W) \right\|^2 + \sum_{\mathbf{m}_{\text{cyl},k}} \left\| \mathbf{r}_{\text{cyl}}(\mathbf{m}_{\text{cyl},k}, \mathbf{T}_{L_k}^W) \right\|^2. \quad (7)$$



Fig. 2. Parked Boeing 737-500 in the hangar test environment, iGPS transmitters are mounted on yellow tripods.

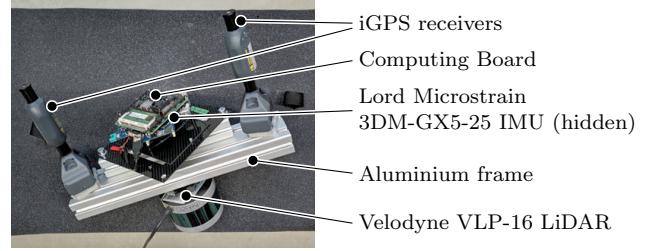


Fig. 3. Hardware setup used for the measurements, consisting of LiDAR, IMU, iGPS receivers and a data logging device.

3. EXPERIMENTAL VALIDATION

The framework was evaluated by several measurement runs in a GNSS-denied hangar environment with the aircraft inspection object depicted in Fig. 2. In order to investigate higher movement dynamics without safety concerns, the measurements were carried out manually using the hardware setup shown in Fig. 3 instead of the UAV. The mounting is equipped with iGPS receivers, that enable millimeter precise positioning using several beacons that were mounted around the test area. This indoor localization system is used as a reference for the LiDAR odometry but since it is an optical system with a very limited field of view, only sparse measurements were available over the duration of the measurement campaign due to frequently interrupted transmitter receiver line-of-sight.

Fig. 4 shows the unfiltered linear acceleration and angular velocity measured by the IMU. The acceleration measurements contain the share of the gravitational vector that is visible in the z -component and the movement comprises high pitch dynamics as the high amplitude oscillation of ω_θ shows. This pitch movement serves the purpose to capture

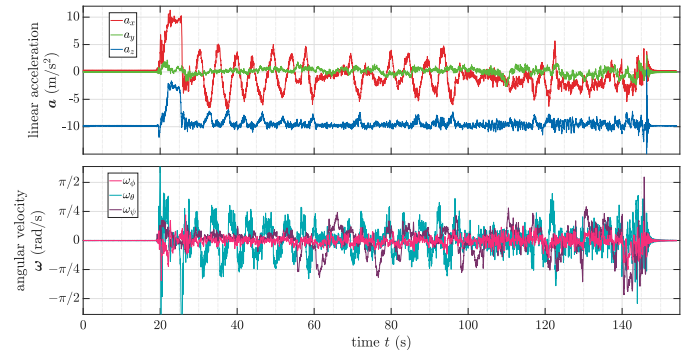


Fig. 4. Linear acceleration and angular velocity IMU measurement data.

the entire height of the aircraft with the LiDAR sensor in spite of the close working distance. At 20s the measurement was initialized with a rapid turn over of the LiDAR sensor by 180° to support precise time synchronization of LiDAR and IMU data in post-processing.

In Fig. 5 the results of the pose estimation by the LiDAR inertial odometry and mapping is shown. By the two optimizations, we obtain plausible, sufficiently smooth position and orientation data over the entire measurement period without any outliers. Most notably, the position drift in z -direction at the end of the measurement duration of 2.5 min is approximately 20 cm, despite the perpetual pitch oscillation and significant yaw rotation. As no loop closing was employed to obtain these results, start and end position of the measurement differed and thus, the drift of the x - and y -component cannot be extracted from the data. Due to their mm-precision, iGPS measurements can be used as ground truth for position in the hangar, when available. To compare the results of the LiDAR inertial SLAM to the iGPS reference however, the unknown rigid 3D transformation between the iGPS coordinate frame and the world frame W of the global map from equation 7 has to be determined. This can be considered a registration problem as well, not of LiDAR scans but of the N sparse iGPS 3D position measurements $\mathbf{p}_k^{\text{iGPS}}, k \in \{1, \dots, N\}$ to the dense odometry position estimates $\mathbf{p}_{B_k}^W$. One possibility to solve this problem is coarse manual alignment, followed by Iterative Closest Point (ICP, Besl and McKay (1992)). However, as the positions are part of trajectories and hence timestamped, point correspondences are already known and there is no necessity for the ICP heuristic. Therefore, the solution for finding the least-squares-optimal transformation can be given in closed form as follows in accordance to Eggert et al. (1997):

- (1) Translate both point clouds so that their respective centroids lie in the origin:

$$\begin{aligned} \tilde{\mathbf{p}}_k^{\text{iGPS}} &= \mathbf{p}_k^{\text{iGPS}} - \bar{\mathbf{p}}^{\text{iGPS}} \quad \forall k \in \{1, \dots, N\} \\ \tilde{\mathbf{p}}_{B_k}^W &= \mathbf{p}_{B_k}^W - \bar{\mathbf{p}}_B^W \quad \forall k \in \{1, \dots, N\} \end{aligned} \quad (8)$$

$$\text{with } \bar{\mathbf{p}}^{\text{iGPS}} = \frac{1}{N} \sum_{k=1}^N \mathbf{p}_k^{\text{iGPS}}, \bar{\mathbf{p}}_B^W = \frac{1}{N} \sum_{k=1}^N \mathbf{p}_{B_k}^W$$

- (2) Construct the point scatter matrices:

$$\begin{aligned} \mathbf{P}^{\text{iGPS}} &= [\tilde{\mathbf{p}}_1^{\text{iGPS}} \quad \tilde{\mathbf{p}}_2^{\text{iGPS}} \quad \dots \quad \tilde{\mathbf{p}}_N^{\text{iGPS}}] \\ \mathbf{P}_B^W &= [\tilde{\mathbf{p}}_{B_1}^W \quad \tilde{\mathbf{p}}_{B_2}^W \quad \dots \quad \tilde{\mathbf{p}}_{B_N}^W] \end{aligned} \quad (9)$$

- (3) The rotation $\mathbf{R}_{\text{iGPS}}^W$ of the optimal rigid transformation between the two coordinate frames can then be obtained by singular value decomposition:

$$\mathbf{R}_{\text{iGPS}}^W = \mathbf{V}\mathbf{U}^T, \text{ with } \mathbf{P}^{\text{iGPS}}\mathbf{P}_B^{WT} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T \quad (10)$$

- (4) Lastly, the translation $\mathbf{t}_{\text{iGPS}}^W$ of the optimal rigid transformation follows with the above result:

$$\mathbf{t}_{\text{iGPS}}^W = \bar{\mathbf{p}}_B^W - \mathbf{R}_{\text{iGPS}}^W \bar{\mathbf{p}}^{\text{iGPS}} \quad (11)$$

The coordinate-transformed iGPS-measurements are indicated as orange circles in Fig. 5, proving that the positions from the pose estimates of the LiDAR inertial odometry and mapping are valid. The RMS-value of the position errors is 25.0 cm and their arithmetic mean 21.2 cm.

For defect map generation however, the mapping result is more relevant than the state estimation. In Fig. 6 the

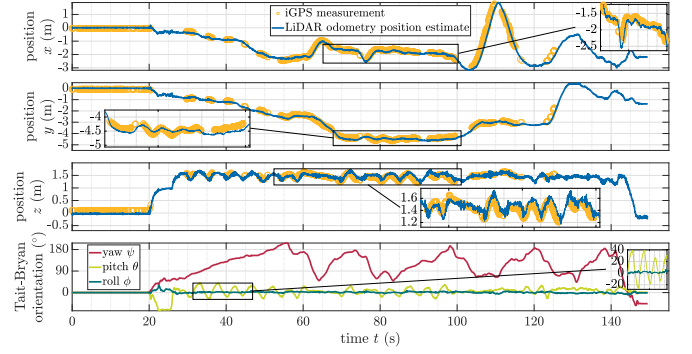


Fig. 5. Position and heading of the LiDAR estimated poses and comparison to coordinate-transformed iGPS measurements.

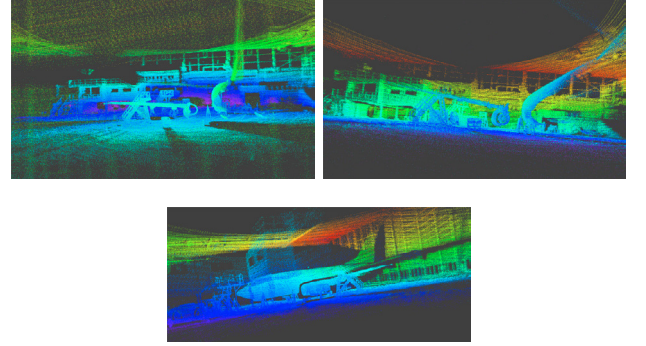


Fig. 6. Resulting point cloud map from measurement 1.

registration result is given with color coded z -coordinate. But the 3D map is most useful, when visualizing the LiDAR measured intensity of the returns: points are colored using grayscale, where white color (value 255 in 8-bit brightness representation) equals maximum intensity. The results are shown in Fig. 7. As stated initially, the depicted 2D-projections of the map from a variety of angles display a level of detail that is remotely comparable to single-channel camera images plus the benefit of precise 3D referencing of each point, which is an unexpected result for a UAV-suitable 16-layer laserscanner. In Fig. 8 the mapping results are given for a second test run with a different trajectory. Contrast was enhanced for better visualization of the few remaining blind spots of the map that could be compensated by more thorough scanning. All mapping results in Fig. 6-8 were created solely using the SLAM approach without employing the external positioning data. In future work, the resulting 3D maps could be further improved by including loop closure and should be merged with high resolution camera pictures and CNN-based image classification for convenient defect localization.

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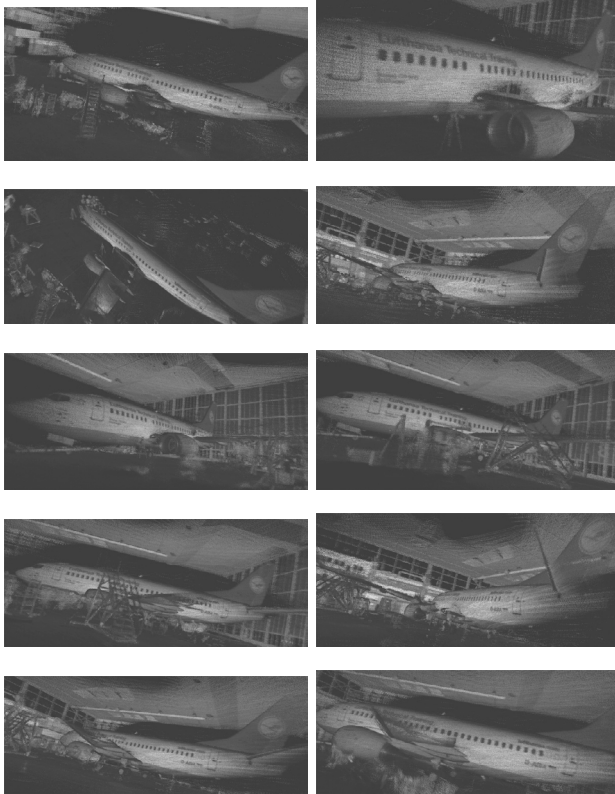


Fig. 7. High resolution 3D point cloud mapping result from measurement 1 with grayscale-coded intensity.

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Fig. 8. High resolution 3D point cloud mapping result from measurement 2 with grayscale-coded intensity.

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